

# Numerical Analysis (SCI19 3111) — Final Cheat Sheet

Regression · Integration · ODEs · Linear Algebra (+ midterm recap)

## 1 — Regression / Data Fitting (Th.1–6)

**Regression  $\neq$  interpolation:** interpolation trusts every measurement and passes through ALL points; regression assumes noise and finds the best OVERALL trend, passing through (almost) none. Deviations  $d_i = y_i - \phi(x_i)$ , data  $(x_i, y_i)$ ,  $i = 0..n$  ( $n+1$  points). **Error choices:** naive  $S = \sum d_i$  (rejected:  $+$ ,  $-$  cancel); abs  $S = \sum |d_i|$  (rejected: kink at 0, no derivative); **squared**  $S = \sum d_i^2$  (chosen: smooth everywhere, so “set gradient = 0” gives solvable linear equations; big misses punished quadratically). Example (Lec14): pts  $(1, 2), (2, 1)$ , line through  $(1, 1), (2, 2)$ :  $d_0 = 1, d_1 = -1$ : naive  $S = 0$  (cancels!), abs  $S = 2$ , squared  $S = 2$ .

### Linear regression

$\hat{y} = a_0 + a_1x$ ,  $S = \sum (y_i - a_0 - a_1x_i)^2$ . General fit  $\phi(x, a_0..a_\ell)$ :  $\frac{\partial S}{\partial a_k} = -2 \sum_{i=0}^n (y_i - \phi(\dots)) \frac{\partial \phi}{\partial a_k} = 0$ ,  $k = 0..\ell$  ( $\ell+1$  equations). For the line this gives the two normal equations below.  $\partial_{a_0}$ :  $\sum y_i - (n+1)a_0 - a_1 \sum x_i = 0$ ;  $\partial_{a_1}$ :  $\sum x_i y_i - a_0 \sum x_i - a_1 \sum x_i^2 = 0$ . Divide by  $n+1 \Rightarrow a_0 + a_1 \bar{x} = \bar{y}$ ,  $a_0 \bar{x} + a_1 \bar{x}^2 = \overline{xy}$ ; Cramer gives

$$a_1 = \frac{\overline{xy} - \bar{x}\bar{y}}{\bar{x}^2 - \bar{x} \cdot \bar{x}}, \quad a_0 = \bar{y} - a_1 \bar{x}$$

with  $\bar{x} = \frac{1}{n+1} \sum x_i$ ,  $\bar{x}^2 = \frac{1}{n+1} \sum x_i^2$ ,  $\overline{xy} = \frac{1}{n+1} \sum x_i y_i$ .

**Trap:**  $x^2 \neq \bar{x}^2$ ; denominator = 0 iff all  $x_i$  equal (no fit possible).

### Hessian + minimum test

$$H = \begin{pmatrix} S_{a_0 a_0} & S_{a_0 a_1} \\ S_{a_1 a_0} & S_{a_1 a_1} \end{pmatrix}$$

$$S_{a_0 a_0} = 2(n+1), \quad S_{a_0 a_1} = 2 \sum x_i, \quad S_{a_1 a_1} = 2 \sum x_i^2$$

$$\det H = 4 \left[ (n+1) \sum x_i^2 - \left( \sum x_i \right)^2 \right]; \min \iff S_{a_0 a_0} > 0 \text{ AND } \det H > 0$$

Intuition:  $H$  = curvature of the error bowl; positive-definite = exactly one bottom.

**Cauchy–Schwarz**  $\Rightarrow \det H \geq 0$ ;  $> 0$  if  $\geq 2$  distinct  $x_i \Rightarrow$  unique global min.

### Multivariable + pseudoinverse

$\hat{y} = a_0 + a_1x_1 + \dots + a_mx_m$ ,  $X$  is  $(n+1) \times (m+1)$  with first col of 1's.  $S = \|\bar{y} - X\bar{a}\|^2$ .

$$X^T X \bar{a} = X^T \bar{y}, \quad \bar{a} = \underbrace{(X^T X)^{-1} X^T}_{X^+} \bar{y}$$

$X^+ =$  Moore–Penrose pseudoinverse ( $= X^{-1}$  when  $X$  square). Idea: projects  $\bar{y}$  onto the column space of  $X$ . **Trap:**  $X^+ X = I$  holds;  $XX^+ = I$  generally does NOT.

### ML: classification

Features = inputs  $(x_1, x_2)$ ; plane  $a_0 + a_1x_1 + a_2x_2 = 0 \cap x_1x_2$ -plane = **decision boundary** (class 1 vs 2). Logistic  $\sigma(z) = \frac{1}{1+e^{-z}} : \mathbb{R} \rightarrow [0, 1]$ ; classifier  $\sigma(a_0 + a_1x_1 + a_2x_2)$ .

## 2 — Linearization

**Goal:** transform  $(\hat{x}, \hat{y})$  so model is linear; fit; inverse-transform back. Logs turn  $\times$  into  $+$  and powers into  $\times$ . Steps: (1) change vars; (2) fit  $\hat{y} = a_0 + a_1\hat{x}$ ; (3) recover  $A, B$ ; (4) check fit.

- $y = Ae^{Bx}$ :  $\hat{x} = x$ ,  $\hat{y} = \ln y$ ;  $a_0 = \ln A$ ,  $a_1 = B \iff A = e^{a_0}$ .
- $y = Ax^n$ :  $\hat{x} = \ln x$ ,  $\hat{y} = \ln y$ ;  $a_0 = \ln A$ ,  $a_1 = n$ .
- $y = A + \frac{B}{x}$ :  $\hat{x} = \frac{1}{x}$ ,  $\hat{y} = y$ ;  $a_0 = A$ ,  $a_1 = B$ .
- $y = \frac{1}{A+Bx}$ :  $\hat{x} = x$ ,  $\hat{y} = \frac{1}{y}$ ;  $a_0 = A$ ,  $a_1 = B$ .
- $y = \frac{Ax}{B+x}$ :  $\hat{x} = \frac{1}{x}$ ,  $\hat{y} = \frac{1}{y}$ ;  $a_0 = \frac{1}{A}$ ,  $a_1 = \frac{B}{A} \iff A = \frac{1}{a_0}$ ,  $B = \frac{a_1}{a_0}$ .

**Trap:** not everything linearizes (e.g.  $a_0e^{-x} \sin(a_1 +$

$2\pi x)$ ). **Model from graph:** straight  $\rightarrow$  linear; explosive growth  $\rightarrow Ae^{Bx}$ ; power curve  $\rightarrow Ax^n$ ; saturating hyperbola  $\rightarrow Ax/(B+x)$ ; reciprocal decay  $\rightarrow A+B/x$ .

## 3 — Grids + Riemann Sums

Partition  $a = x_0 < \dots < x_n = b$ ,  $h_i = x_i - x_{i-1}$ , sample  $x_i^* \in [x_{i-1}, x_i]$ :  $\int_a^b f \approx \sum f(x_i^*)h_i$ .

**Uniform:**  $h_i \equiv h = \frac{b-a}{n}$ ,  $n+1$  points.

**Non-uniform:** varying  $h_i$  (adaptive: fine where  $f$  changes fast, coarse elsewhere). **Riemann integral** =  $\lim_{\max h_j \rightarrow 0} \sum h_i f(x_i^*)$ ;  $f$  bounded, at most countably many jumps.

## 4 — Rectangle / Trapezoid / Simpson

**Sample point:**  $x_i^* = \alpha x_{i-1} + (1-\alpha)x_i$ :  $\alpha = 1$  left,  $\alpha = 0$  right,  $\alpha = \frac{1}{2}$  midpoint. (Quiz7:  $0.2x_{i-1} + 0.8x_i$  is  $\alpha = 0.2$ ; Lec16:  $x_{i-1} + h_i/3$  is  $\alpha = \frac{2}{3}$ ,  $x_{i-1} + 2h_i/3$  is  $\alpha = \frac{1}{3}$ .) General non-uniform: left  $\sum h_i f(x_{i-1})$ , right  $\sum h_i f(x_i)$ , mid  $\sum h_i f(\frac{x_{i-1}+x_i}{2})$ .

$$\tilde{I}_{\text{Tr}} = \frac{1}{2} \sum_{i=1}^n (f(x_{i-1}) + f(x_i)) h_i$$

Uniform trapezoid:  $\frac{h(f_0+f_n)}{2} + h \sum_{i=1}^{n-1} f_i$ . **Trap idea:** one strip = rectangle  $hQ_1$  + triangle  $\frac{h}{2}(Q_2 - Q_1) = h(Q_1 + Q_2)/2$ ; uniform version averages neighbors (interior  $f_i$  counted twice  $\rightarrow$  single sum). **Simpson** ( $n = 2m$  even, uniform): fits parabola on each pair. Why  $O(h^4)$ ? Parabola matches value+slope+curvature (trap: only value+slope).

$$I_{\text{parab}} = \frac{h}{3} \left\{ f_0 + f_n + 4 \sum_{\text{odd } i} f_i + 2 \sum_{\text{even } i < n} f_i \right\}$$

**Sketch:** rectangles = bars at  $x_i^*$ ; trapezoids = straight segments node-to-node; Simpson = parabola arcs per pair (blank axes: mark  $n+1$  nodes first). Simpson weights: endpoints  $\times 1$ , odd indices  $\times 4$ , even interior  $\times 2$ .

### Accuracy estimates

- left/right rectangle  $O(h)$ ; midpoint  $O(h^2)$ ; trapezoid  $O(h^2)$ ; Simpson  $O(h^4)$ .
- Read  $O(h^p)$  as: halving  $h$  divides the error by  $2^p$  (rect  $\div 2$ , mid/trap  $\div 4$ , Simpson  $\div 16$ ).
- Left-rect global:  $|E| \leq \frac{M_1(b-a)}{2} h$ ,  $M_1 = \max |f'|$ . Midpoint:  $|E| \leq \frac{M_2(b-a)}{24} h^2$  (half the trapezoid constant  $1/12$ ). (Taylor + symmetry kills  $f'$  term.)
- No elementary antiderivative ( $\int e^{x^2} dx$ ,  $\int \frac{dx}{\ln x}$ ): cross-check Simpson vs trapezoid.

## 5 — ODEs: IVP + Numeric Methods

$y' = f(x, y)$  (slope field = direction at each point). General solution  $y(x, C)$  = whole family of curves; IVP (Cauchy)  $+y(x_0) = y_0$  picks the ONE curve through  $(x_0, y_0)$  (fix  $C$  by plugging IC; existence/uniqueness assumed). **Check-solution trap:** differentiate candidate AND plug into ODE — both must hold for ALL  $C$ . **Bridge (Lec19):** Euler comes from the finite difference  $y' \approx \frac{y_{i+1}-y_i}{h}$ : replace  $y'$  in  $y' = f(x, y)$  and solve for  $y_{i+1}$ .

$$y_{i+1} = y_i + hf(x_i, y_i), \quad x_{i+1} = x_i + h \quad (\text{Euler})$$

$$y_{i+1} = y_i + hf(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}}), \quad x_{i+\frac{1}{2}} = x_i + \frac{h}{2}, \quad y_{i+\frac{1}{2}} = y_i + \frac{h}{2} f(x_i, y_i)$$

**RK4:** weighted average of 4 trial slopes with weights  $(1, 2, 2, 1)/6$ :  $k_1 = f(x_n, y_n)$ ;  $k_2 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$ ;  $k_3 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2)$ ;  $k_4 = f(x_n + h, y_n + hk_3)$ .  $y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$ . **Global errors:** Euler  $O(h)$  (local  $O(h^2) \times N \sim 1/h$ ); midpoint  $O(h^2)$ ; RK4  $O(h^4)$  (4 evals/step). **Euler error idea:** Taylor gives local  $O(h^2)$  per step;  $N = (b - x_0)/h = O(1/h)$  steps accumulate to

global  $O(h)$ . **Midpoint idea:** predict midpoint slope with one Euler half-step, then step full  $h$  along it; symmetric  $\Rightarrow$  local  $O(h^3)$ , global  $O(h^2)$ .

## 6 — Linear Algebra

det = oriented area (2D) / volume (3D), up to sign; det  $\neq 0 \iff$  invertible.  $\det(AB) = \det A \det B$ .  $Ax = b$  unique  $\iff \det A \neq 0 \iff n$  pivots.

**Scalar product:**  $\bar{a} \cdot \bar{b} = a^x b^x + a^y b^y = |\bar{a}| |\bar{b}| \cos \alpha$  ( $+a^z b^z$  in 3D);  $|\bar{a}| = \sqrt{(a^x)^2 + (a^y)^2}$ .

**Basic ops:**  $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{pmatrix} = \begin{pmatrix} 17 \\ 39 \end{pmatrix}$  (row  $\cdot$  column).

**Projection (scalar length):**

$\text{proj}_{\bar{b}} \bar{a} = \frac{\bar{a} \cdot \bar{b}}{|\bar{b}|} \frac{\bar{b}}{|\bar{b}|} = |\bar{a}| \cos \alpha$ . Basis:  $\text{proj}_{\bar{b}} \bar{a} = a^x$ ,

$\text{proj}_{\bar{b}} \bar{a} = a^y$ . **Worked:**  $\bar{a} = (0, 1)$ ,  $\bar{b} = (2, 0)$ :

$\text{proj}_{\bar{b}} \bar{a} = 1$ ,  $\text{proj}_{\bar{b}} \bar{b} = 2$ .

**Area:** parallelogram  $S = a^y b^x - a^x b^y = -\det$ ; area =  $|S| = |\det|$ . **Worked:**  $u = (3, 1)$ ,  $v = (1, 2)$ :  $S = 1 \cdot 1 - 3 \cdot 2 = -5$ , area 5. **Volume (3D):**  $|\det|$  of the  $3 \times 3$ ; **Worked:**  $\text{diag}(1, 2, 3)$  volume = 6.

**Minor/cofactor:** Minor( $a_{ij}$ ) = det deleting row  $i$ , col  $j$ ; cofac( $a_{ij}$ ) =  $(-1)^{i+j}$  Minor( $a_{ij}$ ). **Worked:** cofac( $a_{23}$ ) =  $-(a_{11}a_{32} - a_{12}a_{31}) = a_{31}a_{12} - a_{11}a_{32}$ . **Laplace:**  $\det A = \sum_j a_{kj} \text{cofac}(a_{kj})$  (any row  $k$ ).

**Row swap** flips sign. **Worked:**

$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}$ : cofactors 2, 2,  $-3 \Rightarrow \det = 1(2) + 2(2) + 3(-3) = -3$ .

**Triangular:** zeros above (lower) or below (upper) diagonal; det = product of diagonal. **Worked**

**(5 $\times$ 5):**  $\text{diag}(1, 2, 3, 4, 5) \Rightarrow \det = 120 = 5!$ .

**LU:**  $A = LU$  ( $L$  unit-lower,  $U$  upper; set  $\ell_{ii} = 1$ ).

**Worked (2 $\times$ 2):**

$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \begin{bmatrix} 3 & 4 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix}$   
( $u_{22} = 4 - 3 \cdot 2 = -2$ ). **Det:**  $\det A = \det L \cdot \det U$ ; here  $1 \cdot (-2) = -2$  **Check** =  $1 \cdot 4 - 2 \cdot 3$ .

**Worked (3 $\times$ 3):**  $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 7 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 1 \end{bmatrix}$   
( $\ell_{12}=4, u_{41}=3, u_{51}=-6, \ell_{22}=7, \ell_{32}=2, u_{61}=1$ );  
 $\det = 1 \cdot (-3) \cdot 1 = -3$ .

**Solve**  $Ax = y$ :  $Ax = L(Ux) = y$ ; set  $z = Ux$ : solve  $Lz = y$  (forward), then  $Ux = z$  (backward).

**Worked:**  $A$  above,  $y = [1, -1]^T$ :  $z_1 = 1$ ,

$3 + z_2 = -1 \Rightarrow z_2 = -4$ ;  $-2z_2 = -4 \Rightarrow z_2 = 2$ ,

$x_1 + 4 = 1 \Rightarrow x_1 = -3$ .  $\boxed{x = [-3, 2]^T}$  **Check:**

$Ax = [1, -1]^T$ .

## 7 — Midterm Recap (compact)

- Taylor:**  $f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n$ ,  $R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x - x_0)^{n+1}$ ,  $|R_n| \leq \frac{M_{n+1}}{(n+1)!}$ .
- Odd fn: only odd powers (deg 8  $\rightarrow x^7$ ); index  $R_{2m+1}$ , bound  $\frac{12^{2m+2}}{(2m+2)!}$ .
- Maclaurin:**  $e^x, \sin, \cos$  converge  $\forall \mathbb{R}$ ;  $\frac{1}{1-x} = \sum x^k, \ln(1+x), \arctan$  for  $|x| < 1$ .
- Horner:** tableau at  $c$  gives  $p(c)$  (last) + quotient  $q$ ; tableau on  $q$  gives  $p'(c)$ .
- Interpolation:**  $\ell_i(x) = \prod_{j \neq i} \frac{x - x_j}{x_i - x_j}$ ,  $p = \sum y_i \ell_i$ ; err =  $\frac{f^{(n+1)}(\xi)}{(n+1)!} \prod (x - x_i)$  ( $= 0$  at nodes).
- Vandermonde  $\det = \prod_{i < j} (x_j - x_i) \neq 0$  (distinct nodes); DD:  $f[x_i, x_{i+1}] = \frac{\Delta f}{\Delta x}$ , Newton  $p_n = f[x_0] + \sum f[x_0..x_k] \prod_{j < k} (x - x_j)$ .

## 8 — Worked Mini-Examples

### Coffee regression (Quiz 6)

Given  $x : 0, 1, 2, 3$ ;  $y : 3, 5, 8, 7$  ( $n+1=4$  pts). Step 1 — sums (add term by term):  
 $\sum x_i = 0+1+2+3 = 6$ ;  $\sum y_i = 3+5+8+7 = 23$ ;  
 $\sum x_i^2 = 0+1+4+9 = 14$ ;  
 $\sum x_i y_i = 0+5+16+21 = 42$ . Step 2 — means (divide by 4):  $\bar{x} = 6/4 = 1.5$ ;  $\bar{y} = 23/4 = 5.75$ ;  
 $\bar{x}^2 = 14/4 = 3.5$ ;  $\overline{xy} = 42/4 = 10.5$ . Step 3 — coeffs (plug into boxed formula):  $a_1 = \frac{10.5 - 1.5 \cdot 5.75}{3.5 - 1.5^2} = \frac{10.5 - 8.625}{3.5 - 2.25} = \frac{1.875}{1.25} = 1.5$ ;  $a_0 = 5.75 - 1.5 \cdot 1.5 = 5.75 - 2.25 = 3.5$ .  $\boxed{\hat{y} = 3.5 + 1.5x}$ . Step 4 — predict  $x=4$ :  $\hat{y} = 3.5 + 1.5 \cdot 4 = 3.5 + 6 = 9.5$

(mark (4, 9.5) on the scatter plot with the line). **Plot recipe (Quiz6 Q2):** scatter the 4 data points, draw the line through (0, 3.5) and (4, 9.5) — regression line need NOT pass through data points. Step 5 — check fit:  $\hat{y}$  : 3.5, 5.0, 6.5, 8.0 vs  $y$  : 3, 5, 8, 7; residuals  $-0.5, 0, 1.5, -1.0$ ; squared error  $S = 0.25 + 0 + 2.25 + 1 = 3.5$  (this  $S$  is what least squares minimized). **Battery prediction:** at  $t=70$ :  $0.714 \cdot 70 + 3.57 = 53.55\%$  (extrapolation — flag as less reliable). **Hessian:** from Step 1 sums  $(\sum x_i = 6, \sum x_i^2 = 14)$ :  $S_{a_0 a_0} = 2 \cdot 4 = 8$ ,  $S_{a_0 a_1} = 2 \cdot 6 = 12$ ,  $S_{a_1 a_1} = 2 \cdot 14 = 28$ ;  $\det H = 8 \cdot 28 - 12^2 = 224 - 144 = 80 > 0$  ( $+8 > 0$ )  $\Rightarrow$  min.

**Battery fit (Lec 14)**

Given  $t$  : 0, 10, 20, 30, 40, 50, 60; charge 5, 10, 15, 25, 35, 40, 45 ( $n+1=7$ ). Sums:  $\sum t_i = 0+10+20+30+40+50+60 = 210$ ;  $\sum y_i = 5+10+15+25+35+40+45 = 175$ ;  $\sum t_i^2 = 0+100+400+900+1600+2500+3600 = 9100$ ;  $\sum t_i y_i = 0+100+300+750+1400+2000+2700 = 7250$ . Means (divide by 7):  $\bar{x} = 210/7 = 30$ ;  $\bar{y} = 175/7 = 25$ ;  $\bar{x^2} = 9100/7 = 1300$ ;  $\bar{xy} = 7250/7 \approx 1035.71$ . Coeffs:  $a_1 = \frac{1035.71-30 \cdot 25}{1300-30^2} = \frac{1035.71-750}{1300-900} = \frac{285.71}{400} \approx 0.714$ ;  $a_0 = 25 - 0.714285 \cdot 30 = 25 - 21.43 \approx 3.57$ .  $\phi(x) = 0.714x + 3.57$ . Check fit:  $\hat{y}$  : 3.57, 10.71, 17.85, 24.99, 32.13, 39.27, 46.41 vs  $y$  : 5, 10, 15, 25, 35, 40, 45 (residuals 1.43,  $-0.71$ ,  $-2.85$ ,  $0.01$ ,  $2.87$ ,  $0.73$ ,  $-1.41$ ).

**Exp linearization (HW09)**

Given  $x$  : 1, 2, 3, 5,  $y$  : 8, 4, 2, 1; model  $y = Ae^{Bx} \Rightarrow$  transform  $\tilde{x} = x$ ,  $\tilde{y} = \ln y$ . Step 0 — transform each  $y$ :  $\ln 8 \approx 2.07944$ ,  $\ln 4 \approx 1.38629$ ,  $\ln 2 \approx 0.69315$ ,  $\ln 1 = 0$ . So  $\tilde{x}$  : 1, 2, 3, 5;  $\tilde{y}$  : 2.07944, 1.38629, 0.69315, 0. Step 1 — means ( $n+1=4$ ):  $\bar{\tilde{x}} = (1+2+3+5)/4 = 11/4 = 2.75$ ;  $\bar{\tilde{y}} = (2.07944+1.38629+0.69315+0)/4 = 4.15888/4 \approx 1.03972$  (note  $= \ln 64/4 = 1.5 \ln 2$ );  $\bar{\tilde{x}^2} = (1+4+9+25)/4 = 39/4 = 9.75$ ;  $\bar{\tilde{x}\tilde{y}} = (1 \cdot 2.07944 + 2 \cdot 1.38629 + 3 \cdot 0.69315 + 5 \cdot 0)/4 = 6.93147/4 \approx 1.73287$  (note  $= 2.5 \ln 2$ ). Step 2 — fit:  $a_1 = \frac{1.73287 - 2.75 \cdot 1.03972}{9.75 - 2.75^2} = \frac{1.73287 - 2.85923}{9.75 - 7.5625} = \frac{-1.12636}{2.1875} \approx -0.5149$ ;  $a_0 = 1.03972 - (-0.5149) \cdot 2.75 = 1.03972 + 1.41598 \approx 2.4557$ . Step 3 — back-transform ( $A = e^{a_0}$ ,  $B = a_1$ ):  $A = e^{2.4557} \approx 11.65$ ,  $B \approx -0.515 < 0$ . Check:  $\hat{y} = 11.65e^{-0.515x}$  gives 7.0, 4.2, 2.5, 0.9 vs 8, 4, 2, 1 — reasonable. **Nonlinear example (Lec15):**  $x$  : 1, 2, 3, 4,  $y$  : 0.45, 0.81, 1.05, 1.13 for  $y = \frac{Ax}{B+x}$ :  $\tilde{x} = 1/x$ ,  $\tilde{y} = 1/y$  give  $a_1 \approx 1.834$ ,  $a_0 \approx 0.368$ , so  $A = 1/a_0 \approx 2.71$ ,  $B = a_1/a_0 \approx 4.98$ .

**Trapezoid  $\sin x$  (Quiz 7)**

Given  $[0, 3\pi/2]$ , uniform  $h = \pi/4 \Rightarrow n = (3\pi/2)/(\pi/4) = 6$  subintervals, 7 nodes  $x_i = i\pi/4$ . Step 1 — evaluate  $f(x) = \sin x$ :  $\sin 0 = 0$ ;  $\sin \frac{\pi}{4} = \frac{\sqrt{2}}{2} \approx 0.70711$ ;  $\sin \frac{\pi}{2} = 1$ ;  $\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}$ ;  $\sin \pi = 0$ ;  $\sin \frac{5\pi}{4} = -\frac{\sqrt{2}}{2}$ ;  $\sin \frac{3\pi}{2} = -1$ . Step 2 — uniform trapezoid (endpoints half-weighted):  $\tilde{I} = \frac{h}{2}(f_0+f_6) + h \sum_{i=1}^5 f_i$  with  $\sum_{i=1}^5 f_i = \frac{\sqrt{2}}{2} + 1 + \frac{\sqrt{2}}{2} + 0 - \frac{\sqrt{2}}{2} = 1 + \frac{\sqrt{2}}{2} \approx 1.70711$ . Step 3 — multiply out:  $\tilde{I} = \frac{h}{2}(f_0+f_6) + h \sum_{i=1}^5 f_i = \frac{\pi}{4} \left( \frac{1}{2}(-1) + 1.70711 \right) = \frac{\pi}{4} (1.20711) \approx 0.78540 \cdot 1.20711 \approx 0.94806$ . Step 4 — exact  $I = \int_0^{3\pi/2} \sin x \, dx = [-\cos x]_0^{3\pi/2} = -\cos \frac{3\pi}{2} + \cos 0 = 0 + 1 = 1$ ; err  $= |1 - 0.94806| \approx 0.05194$ . **How to draw trapezoids (Pract.6):** (1) mark all  $n+1$  nodes on the axis; (2) plot  $(x_i, f(x_i))$ ; (3) join consecutive points with straight segments; (4) each strip is a trapezoid (drop verticals to the axis). Parabolae (Simpson): same nodes, one arc per pair through its 3 points.

**Simpson  $\int_2^4 \frac{dx}{\ln x}$  (Quiz 8)**

Given  $n = 8$  (even  $\checkmark$ ), 9 points,  $h = (4 - 2)/8 = 0.25$ . Nodes  $x_i = 2 + 0.25i$ :

2, 2.25, 2.5, 2.75, 3, 3.25, 3.5, 3.75, 4. Step 1 — evaluate  $f(x) = 1/\ln x$  at each node:  $f_0 = 1/\ln 2 \approx 1.44270$ ;  $f_1 = 1/\ln 2.25 \approx 1.23315$ ;  $f_2 = 1/\ln 2.5 \approx 1.09136$ ;  $f_3 = 1/\ln 2.75 \approx 0.98853$ ;  $f_4 = 1/\ln 3 \approx 0.91024$ ;  $f_5 = 1/\ln 3.25 \approx 0.84842$ ;  $f_6 = 1/\ln 3.5 \approx 0.79824$ ;  $f_7 = 1/\ln 3.75 \approx 0.75657$ ;  $f_8 = 1/\ln 4 \approx 0.72135$ . Step 2 — weight (ends  $\times 1$ , odd  $i \times 4$ , even  $i \times 2$ ):  $4 \sum_{\text{odd}} = 4(1.23315 + 0.98853 + 0.84842 + 0.75657) = 4(3.82667) = 15.30668$ ;  $2 \sum_{\text{even}} = 2(1.09136 + 0.91024 + 0.79824) = 2(2.79984) = 5.59968$ . Step 3 — assemble:  $I = \frac{0.25}{3} (1.44270 + 15.30668 + 5.59968 + 0.72135) = \frac{0.25}{3} (23.07041) \approx 0.83333 \cdot 23.07041 \approx 1.92253$ .

**Euler + midpoint  $y' = x - 4y$ ,  $y(0) = -3/4$  (Quiz 9)**

$h = 0.2$ ,  $f(x, y) = x - 4y$ . **Euler:**  $f_0 = 0 - 4(-0.75) = 3$ ;  $y_1 = -0.75 + 0.2(3) = -0.15$ ;  $f_1 = 0.2 - 4(-0.15) = 0.8$ ;  $y_2 = -0.15 + 0.2(0.8) = 0.01$ ;  $f_2 = 0.4 - 0.04 = 0.36$ ;  $y_3 = 0.01 + 0.2(0.36) = 0.082$ . **Midpoint:** half-step  $(0.1, -0.75 + 0.1(3)) = (0.1, -0.45)$ ,  $f = 0.1 + 1.8 = 1.9$   $\Rightarrow y_1 = -0.75 + 0.2(1.9) = -0.37$ ; half-step  $(0.3, -0.37 + 0.1(1.68)) = (0.3, -0.202)$ ,  $f = 0.3 + 0.808 = 1.108$   $\Rightarrow y_2 = -0.37 + 0.2(1.108) = -0.1484$ . Exact  $y(0.4) \approx -0.101304$  (given in quiz; from closed form  $y = x/4 - 1/16 - \frac{11}{16}e^{-4x}$ ): Euler err  $= |-0.101304 - 0.01| \approx 0.11130$ ; midpoint err  $= |-0.101304 - (-0.1484)| \approx 0.04710$ . **IVP from**

**general sol:**  $y = -\frac{1}{4} + \frac{x}{2} + Ce^{-2x}$ ,  $(x_0, y_0) = (\frac{1}{2}, \frac{1}{4})$ :  $\frac{1}{4} = -\frac{1}{4} + \frac{1}{4} + Ce^{-1} \Rightarrow C = e/4 \approx 0.6796$ ,  $y^* = -\frac{1}{4} + \frac{x}{2} + \frac{e}{4}e^{-2x}$ . **HW10 Q2 pattern:**  $y' = \frac{y}{x} + 2x^2$ , (1, 4), general  $y = (x^2 + C)x$ :  $4 = (1 + C) \cdot 1 \Rightarrow C = 3$ ,  $y^* = (x^2 + 3)x$ . Euler pattern:  $x_1 = 1 + h$ ,  $y_1 = 4 + h(4/1 + 2) = 4 + 6h$  (since  $f(1, 4) = 6$ ). Midpoint pattern:  $x_{1/2} = 1 + h/2$ ,  $y_{1/2} = 4 + (h/2) \cdot 6 = 4 + 3h$ . **Check trap:**  $y = Ce^x - 2 \cos 2x - 2 \sin 2x$  is *not* a sol of  $y' = y + 5 \sin 2x$ :  $y' = Ce^x + 4 \sin 2x - 4 \cos 2x$ , but  $y + 5 \sin 2x = Ce^x - 2 \cos 2x + 3 \sin 2x$ ; difference  $= \sin 2x - 2 \cos 2x \neq 0$ . But  $y = \frac{1}{2}(\cos x - \sin x) + Ce^x$  is a sol of  $y' = y - \cos x$ :  $y' = \frac{1}{2}(-\sin x - \cos x) + Ce^x = y - \cos x$ .

**Rule:** differentiate candidate AND plug into ODE — both must hold  $\forall C$ . **Decision boundary**

**mini-example:** plane  $-1 + x_1 + x_2 = 0$ : point (0, 0) gives  $-1 < 0$  (class 1), (1, 1) gives  $+1 > 0$  (class 2);  $\sigma(-1 + x_1 + x_2)$  maps these to  $\approx 0.27$  and  $\approx 0.73$ . **RK4 schemes (no numbers needed):** for  $y' = y + x$ :  $k_1 = y_i + x_i$ ,  $k_2 = y_i + \frac{h}{2}k_1 + x_i + \frac{h}{2}$ ,  $k_3 = y_i + \frac{h}{2}k_2 + x_i + \frac{h}{2}$ ,  $k_4 = y_i + hk_3 + x_i + h$ . For  $y' = y + 1/x$  ( $h=1/12$ ):  $k_1 = x_i y_i + 1/x_i$ ;  $k_2 = (x_i + \frac{1}{24})(y_i + \frac{k_1}{4}) + 1/(x_i + \frac{1}{24})$ ;  $k_3$  same with  $k_2$ ;  $k_4 = (x_i + \frac{1}{12})(y_i + \frac{k_3}{2}) + 1/(x_i + \frac{1}{12})$ .

**Rectangle lattice nodes (Quiz 7)**

Given  $x_k = \frac{3 \ln(1+k/5)}{\ln 2} - 1$ ,  $k = 0..5$ . Sample:  $x_1 = \frac{3 \ln 1.2}{\ln 2} - 1 \approx -0.21090$  (since  $3 \ln 1.2 \approx 0.54696$ ,  $\ln 2 \approx 0.69315$ );  $x_0 = \frac{3 \ln 1}{\ln 2} - 1 = -1$ ;  $x_5 = \frac{3 \ln 2}{\ln 2} - 1 = 2$ . Full list:  $-1.00000, -0.21090, 0.45628, 1.03422, 1.54399, 2.00000$ .  $x_i^* = 0.2x_{i-1} + 0.8x_i$ :  $-0.36872, 0.32285, 0.91863, 1.44204, 1.90880$ . Rect  $i$  has width  $h_i = x_i - x_{i-1}$ , height  $f(x_i^*)$  ( $f = x^3 - 1.25x^2$ ). **How to draw (Pract.5):** (1) mark the  $n+1$  nodes on the  $x$ -axis; (2) for each subinterval compute  $x_i^*$  via the  $\alpha$ -formula (any  $\alpha$ :  $1/3, 2/3, 0.2$ , midpoint); (3) go up to the curve at  $f(x_i^*)$  — that height is the bar top; (4) draw the bar over  $[x_{i-1}, x_i]$  (works below the axis too: height negative  $\Rightarrow$  signed area).

**Accuracy table  $\int_0^1 x^2 dx$  (Lec17,  $h = 0.25$ )**

Nodes  $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$ ;  $f(x) = x^2$  gives  $0, \frac{1}{16}, \frac{1}{4}, \frac{9}{16}, 1$ . Left  $= \frac{1}{64}$ ? **No:**  $f_3 = (\frac{3}{4})^2 = \frac{9}{16}$ , not  $\frac{3}{4}$ . Correct left uses  $f_{0..f_3} = \frac{1}{4}(0 + \frac{1}{16} + \frac{1}{4} + \frac{9}{16}) = \frac{1}{4}(\frac{14}{16}) = \frac{7}{32} \approx 0.21875$ , err  $= |0.33333 - 0.21875| \approx 0.11458$ .

Mid: midpoints  $\frac{1}{8}, \frac{3}{8}, \frac{5}{8}, \frac{7}{8}$ ,  $f^* = \frac{1}{64}, \frac{9}{64}, \frac{25}{64}, \frac{49}{64}$ ;  $\tilde{I} = \frac{1}{4} \cdot \frac{84}{64} \approx 0.32813$  (err 0.00521). Right  $= \frac{1}{4}(\frac{1}{16} + \frac{1}{4} + \frac{9}{16} + 1) = \frac{15}{32} \approx 0.46875$  (err 0.13542). Trap  $= \frac{11}{32} \approx 0.34375$  (err 0.01042). **Derivation idea (Th.10):** Taylor  $f(x) = f(x_{i-1}) + f'(\xi_x)(x - x_{i-1})$ ; integrate over  $[x_{i-1}, x_i]$ : local err  $|E_i| \leq M_1 h_i^2/2$ ; sum:  $|E| \leq \frac{M_1}{2} \sum h_i^2$ ; uniform:  $\sum h_i^2 = nh^2$ ,  $nh = b - a \Rightarrow |E| \leq \frac{M_1(b-a)}{2}h = O(h)$ . Midpoint: expand at  $c_i$ ; odd term  $\int (x - c_i) = 0$  vanishes  $\Rightarrow O(h^2)$ .

**Simpson  $\int_1^2 e^{x^2} dx$  (Lec19,  $n = 4$ )**

Given  $h = (2 - 1)/4 = 0.25$ , nodes  $x_i = 1 + 0.25i$ :  $1, \frac{5}{4}, \frac{3}{2}, \frac{7}{4}, 2$ . Step 1 — evaluate  $f(x) = e^{x^2}$  (square  $x$  first, then exponentiate):  $f_0 = e^1 \approx 2.71828$ ;  $f_1 = e^{(5/4)^2} = e^{1.5625} \approx 4.77073$ ;  $f_2 = e^{(3/2)^2} = e^{2.25} \approx 9.48774$ ;  $f_3 = e^{(7/4)^2} = e^{3.0625} \approx 21.38094$ ;  $f_4 = e^4 \approx 54.59815$ .  $I = \frac{0.25}{3} (2.71828 + 4(26.15167) + 2(9.48774) + 54.59815) \approx 15.0749$ . Trap check: two double-intervals  $T_1 = h(f_0 + f_2) = 0.25(2.71828 + 9.48774) = 0.25(12.20602) \approx 3.0515$ ,  $T_2 = h(f_2 + f_4) = 0.25(9.48774 + 54.59815) = 0.25(64.08589) \approx 16.0215$ , total  $\approx 19.0730 \rightarrow 19.07$  — same ballpark  $\Rightarrow$  likely right. **Simpson sketch (Quiz8 Q1):**  $[0, 240]$ , 6 subintervals  $\Rightarrow h = 40$ , nodes  $0, 40, \dots, 240$ ; one parabola arc per pair of subintervals (3 arcs total); mark nodes, draw each arc through its 3 points.

**Pseudoinverse  $\rightarrow$  1D formulas (HW09)**

$X^T X \bar{a} = X^T \bar{y}$  with  $X = [1 \ x_i]$  gives  $(\sum_{i=1}^{n+1} x_i) \begin{pmatrix} a_0 \\ a_1 \end{pmatrix} = (\sum_{i=1}^{n+1} y_i)$ ; dividing by  $n+1$  and solving (Cramer) returns  $a_1 = \frac{\bar{xy} - \bar{x}\bar{y}}{\bar{x^2} - \bar{x}^2}$ ,  $a_0 = \bar{y} - a_1 \bar{x}$ . Cramer in full:  $\Delta = (n+1) \sum x_i^2 - (\sum x_i)^2$ ;  $a_0 = \frac{(\sum y_i)(\sum x_i^2) - (\sum x_i)(\sum x_i y_i)}{\Delta}$ ;  $a_1 = \frac{(n+1)(\sum x_i y_i) - (\sum x_i)(\sum y_i)}{\Delta}$ ; dividing top and bottom by  $(n+1)^2$  gives the averaged formulas. **HW09 Q2 setup ( $m=2, n=2$ , 3 pts):**  $X$  is  $3 \times 3$  (first col of 1's, then  $x_1^{(i)}, x_2^{(i)}$  rows);  $X^+ = (X^T X)^{-1} X^T$  is  $3 \times 3$ . **Q3:**  $X^+ X = I$  always;  $XX^+ = I$  only if  $X$  square nonsingular.

**Euler  $y' = y + x$  (Lec20, corrected)**

$(x_0, y_0) = (0, 1)$ ,  $h = 0.25$ :  $y_1 = 1 + 0.25(1) = 1.25$ ;  $y_2 = 1.25 + 0.25(1.5) = 1.625$ ;  $y_3 = 1.625 + 0.25(2.125) = 2.15625$ . (Compiled 1.5625/2.078125 used  $f_1 = 1.25$  instead of  $0.25 + 1.25 = 1.5$  — knock-on slip.) Exact  $y^* = 2e^x - x - 1$ :  $y(0.5) \approx 1.79744$ ,  $y(0.75) \approx 2.48400$  — Euler visibly below.

**Midpoint on same ODE:**

$(0.125, 1.125) \Rightarrow y_1 = 1 + 0.25(1.25) = 1.3125$  vs exact 1.31805 — much closer. **Separable pattern:**  $y' = x/y \Rightarrow y \, dy = x \, dx \Rightarrow y^2 = x^2 + C_1$ . Circle IVP  $y' = -x/y$ , (2, 2):  $y^2/2 + x^2/2 = C = 4$  (radius 2).

**IVP slip:**  $y' = y + x$ , (1, 4):  $4 = Ce - 2 \Rightarrow C = 6/e \approx 2.207$  (compiled 2/e forgot +4: check  $y(1) = 6 - 2 = 4 \checkmark$ ).

**Trapezoid, non-uniform lattice**

Given nodes 0, 0.25, 1  $\Rightarrow$  widths  $h_1 = 0.25 - 0 = 0.25$ ,  $h_2 = 1 - 0.25 = 0.75$ ;  $f = x^2$  gives  $f_0 = 0$ ,  $f_1 = 0.0625$ ,  $f_2 = 1$ .  $\tilde{I} = \frac{1}{2} [(f_0+f_1)h_1 + (f_1+f_2)h_2] = \frac{1}{2} [(0+0.0625)(0.25) + (0.0625+1)(0.75)] = \frac{1}{2} (0.015625 + 0.796875) = \frac{1}{2} (0.8125) = 0.40625$ . Exact  $\frac{1}{3}$ ; err  $= |0.33333 - 0.40625| \approx 0.07292$ .

**Laplace expansion**

Given  $A = [[1, 2, 3], [4, 5, 6], [7, 8, 10]]$ , expand along row 1. Step 1 — cofactors:  $\text{cofac}(a_{11}) = +(5 \cdot 10 - 6 \cdot 8) = 50 - 48 = 2$ ;  $\text{cofac}(a_{12}) = -(4 \cdot 10 - 7 \cdot 6) = -(40 - 42) = 2$ ;  $\text{cofac}(a_{13}) = +(4 \cdot 8 - 5 \cdot 7) = 32 - 35 = -3$ . Step 2 — assemble:  $\det A = 1(2) + 2(2) + 3(-3) = 2 + 4 - 9 = -3$  (matches LU  $\det = 1 \cdot (-3) \cdot 1$ ).